Case 2. The walker $s_u(t) > 0$ and we receive a *different* example (so $u \neq v$). Whether s_u decrements in this case is complex to determine, and depends on the precise coordinates ω_u and ω_v. We treat this issue coarsely by modeling the probability of decrement through an average case approximation: if $s_v > 0$, we assume that s_u will always decrement; if $s_v < 0$, we assume that s_u will decrement with some mean probability μ. Since $p(s_v > 0) = \frac{n^+-1}{n-1}$ and $p(s_v < 0) = \frac{n^-}{n-1}$, and the probability of selecting a *different* example overall remains $2/3$, we summarize this case as

$$p(s_u \leftarrow s_u - 1 \,|\, s_u > 0) = \frac{2}{3}\left(\frac{n^+-1}{n-1} + \frac{n^-}{n-1}\mu\right). \tag{C.6}$$

This average case approximation is similar in flavor to the mean field *ansatz* common in physics, and we employ it for similar reasons: it simplifies a complex many-bodied interaction into a simple interaction between a single body and an average field. We validate the accuracy of this approximation later below.

Case 3. The walker $s_u(t) < 0$ and we receive a *same* example. No updates occur in this case. For completeness, we summarize it as

$$p(s_u \leftarrow s_u + 2 \,|\, s_u < 0) = 0. \tag{C.7}$$

Case 4. The walker $s_u(t) < 0$ and we receive a *different* example. We again apply a coarse, average case approximation to model the probability of decrementing. If $s_v < 0$, we assume that s_u will never decrement. If $s_v > 0$, the probability of decrementing again depends on our mean quantity μ. The probability of selecting a *different* example overall remain $2/3$, so we summarize this case as

$$p(s_u \leftarrow s_u - 1 \,|\, s_u < 0) = \frac{2}{3}\left(\frac{n^+}{n-1}\mu\right). \tag{C.8}$$

To gain greater insight into μ, we consider how much a walker's position may drift as it encounters different training examples. Define a walker's *expected drift* to be the quantity $\Delta s(t) = \mathbb{E}[s(t+1) - s(t)]$, averaged over possible walker states s. Then under Eq (C.5) and Eq (C.6), considering a positive walker $s^+ > 0$

$$\begin{aligned}\Delta s^+(t) &= 2\,p(s^+ > 0)\,p(s^+ \leftarrow s^+ + 2 \,|\, s^+ > 0) \\ &\quad - p(s^+ > 0)p(s^+ \leftarrow s^+ - 1 \,|\, s^+ > 0) \\ &= \frac{2n^+}{3n}\left(1 - \frac{n^+ + n^-\mu - 1}{n-1}\right) \\ &= \frac{2n^+n^-(1-\mu)}{3n(n-1)}.\end{aligned} \tag{C.9}$$

Similarly, under Eq (C.7) and Eq (C.8), a negative walker s^- has expected drift

$$\begin{aligned}\Delta s^-(t) &= -\,p(s^- < 0)\,p(s^- \leftarrow s^- - 1 \,|\, s^- < 0) \\ &= -\frac{2n^-n^+\mu}{3n(n-1)}.\end{aligned} \tag{C.10}$$

Suppose $\mu = 1$. In this case, if we encounter a *different* example $(\mathbf{z}_u, \mathbf{z}_v)$ such that $s_u > 0$ and $s_v < 0$, then s_u will always decrement. On average, $\Delta s_u = 0$ while Δs_v is a negative quantity, indicating that s_u will on average remain around the same position while s_v decreases. However, if s_v decreases without bound, there comes a point where $\omega_u + \omega_v < 0$, preventing further decrements. This situation implies that $\mu = 0$, resulting in $\Delta s_u > 0$ and $\Delta s_v = 0$. However, if s_u now increases without bound, there comes a point where $\omega_u + \omega_v > 0$, allowing again further decrements, raising our mean update probability back to $\mu = 1$.

In general, if $|\Delta s_u| < |\Delta s_v|$, we experience further increments until $|\Delta s_u| > |\Delta s_v|$, at which point we experience further decrements, returning us back to $|\Delta s_u| < |\Delta s_v|$. Over a long time period, we might therefore expect our dynamics to settle around an average point $|\Delta s_u| = |\Delta s_v|$. If this is true, then we employ the relation $|\Delta s^+| = |\Delta s^-|$ as a self-consistency condition to solve for μ. Equating (C.9) and (C.10) reveals that $\mu = \frac{1}{2}$.

Altogether, we arrive at the following picture of the walkers' dynamics. Walkers s_u at a positive position drift with an average rate

$$\Delta s_u = \frac{n^+n^-}{3n(n-1)}. \tag{C.11}$$

Meanwhile, walkers s_v at a negative position drift with an average rate $\Delta s_v = -\Delta s_u$. Over long time periods, $s_u \to \infty$ while $s_v \to -\infty$. Because positive updates increment both coordinates ω_u^1 and ω_u^2 equally, we have that $\omega_u^1 \approx \omega_u^2 > 0$. Meanwhile, because negative updates have a higher chance of decrementing a more positive coordinate, we also have that $\omega_v^1 \approx \omega_v^2 < 0$. In this way, we must have overall that $||\mathbf{v}_1|| = ||\mathbf{v}_2||$ and $\mathbf{v}^1 \cdot \mathbf{v}^2/(\left|\left|\mathbf{v}^1\right|\right| \left|\left|\mathbf{v}^2\right|\right|) = 1$ at long time scales, confirming that a weight vector aligned with *same* examples adopts parallel components.

To validate our key assumption on μ, we simulate the Markov process 100 times with $L = 16$ and a batch size of 512. We empirically find that $\mu = 0.508 \pm 0.056$ (given by two standard deviations), matching closely our conjecture that $\mu = \frac{1}{2}$.

One caveat we have not addressed is the case where $n^- = n$. In this case, no further updates occur and the weights are frozen in their current position. However, as we suggest later in Section C.4.3, the corresponding readout of these weights will be relatively small, reducing its impact. In practice, "dead" weights like these that are negatively aligned with all training symbols appear to be rare in trained models.

C.4.2 *Different* case For weights $\mathbf{w}$ corresponding to a readout $a < 0$, similar rules hold but now with flipped signs. Considering again a random walker with position $s_u(t) = \omega_u^1(t) + \omega_u^2(t)$, the following rules govern the walker's dynamics:

1. If $s_u(t) > 0$ and the model receives a training example $(\mathbf{z}_u, \mathbf{z}_u)$, then $s_u(t+1) = s_u(t) - 2$.

2. If the model receives a training example $(\mathbf{z}_u, \mathbf{z}_v)$ or its reverse $(\mathbf{z}_v, \mathbf{z}_u)$, where $u \neq v$ then $s_u(t+1) \leq s(t)+1$.

The rules follow from the update dynamics of the Markov process precisely as before, now for weights sensitive to *different* examples. Note, there is no equivalence to Rule 3 in this case, since a walker may (in most cases) continue to receive either positive or negative updates regardless of the sign of its position. Indeed, this added symmetry to the *different* case simplifies the analysis somewhat compared to the *same* case, where it was necessary to study the interactions between two ensembles of random walkers that evolve in different ways. Here, we may treat all walkers uniformly.

Our general approach for analyzing this case is the same. Conditioned on training examples containing the uth training symbol, recall that observing a *same* pair $(\mathbf{z}_u, \mathbf{z}_u)$ has probability 1/3, and observing a *different* pair $(\mathbf{z}_u, \mathbf{z}_v)$ has probability 2/3. Then we have the following cases:

Case 1. The walker $s_u(t) > 0$ and receives a *same* example. In this case, $s_u(t+1) = s_u(t) - 2$. The probability of observing a *same* pair containing $\mathbf{z}_u$ is 1/3, so we summarize this case as

$$p(s_u \leftarrow s_u - 2 \,|\, s_u > 0) = \frac{1}{3}. \tag{C.12}$$

Case 2. The walker $s_u(t) > 0$ and we receive a *different* example. Whether s_u increments is complex to determine, and depends on the precise coordinates ω_u and ω_v. As before, we treat this issue coarsely by approximating the probability of incrementing through an average case parameter μ. The probability of selecting a *different* example overall remains $2/3$, so we summarize this case as

$$p(s_u \leftarrow s_u + 1 \,|\, s_u > 0) = \frac{2}{3}\mu. \tag{C.13}$$

Case 3. The walker $s_u(t) < 0$ and we receive a *same* example. No updates occur in this case. For completeness, we summarize it as

$$p(s_u \leftarrow s_u - 2 \,|\, s_u < 0) = 0. \tag{C.14}$$

Case 4. The walker $s_u(t) < 0$ and we receive a *different* example. We again use our average case parameter to describe the probability of incrementing. The probability of selecting a *different* example overall remains $2/3$, so we summarize this case as

$$p(s_u \leftarrow s_u + 1 \,|\, s_u < 0) = \frac{2}{3}\mu. \tag{C.15}$$

To obtain a self-consistent condition for μ, we consider again the expected drift of walkers at positive or negative positions. After t timesteps have elapsed, suppose the number of walkers with position $s^+ > 0$ is n^+, and the number of walkers with position $s^- < 0$ is n^-, where $L = n^+ + n^- \equiv n$. Then combining Eq (C.12) and (C.13), the expected drift for positive walkers is

$$\Delta s^+ = \frac{2n^+}{3n}(\mu - 1). \tag{C.16}$$

Combining Eq (C.14) and (C.15), the expected drift of the negative walkers is

$$\Delta s^- = \frac{2n^-}{3n}\mu \tag{C.17}$$

Note, unlike the *same* case, the expected drift for positive walkers is *negative*, and the expected drift for negative walkers is *positive*. Hence, for $\mu \in (0,1)$, if $|\Delta s^+| > |\Delta s^-|$, the number of positive walkers decreases faster than it increases, so we eventually reach a point where $|\Delta s^+| \leq |\Delta s^+|$. However, when $|\Delta s^+| < |\Delta s^-|$, the number of negative walkers decreases faster than it increases, so we oscillate back to $|\Delta s^+| \geq |\Delta s^-|$. Over a long time period, we assume our walkers settle around an average point $|\Delta s^+| = |\Delta s^-|$. If this is true, then as before, we employ the relation $|\Delta s^+| = |\Delta s^-|$ as a self-consistency condition to solve for μ. Equating (C.16) and (C.17) indicates that $\mu = \frac{n^+}{n}$.

There are three potential settings for n^+ to consider: $0 < n^+ < n$, $n^+ = n$, or $n^+ = 0$. Let us begin with $0 < n^+ < n$. Because $n^+ < n$, the average position of positive walkers experiences a net negative drift, gradually bringing them closer to zero. Because $n^+ > 0$, the average position of negative walkers experiences a net positive drift, gradually bringing them closer to zero also. Over a long period of time, we would therefore expect $n^+ \approx n^-$ (and $\mu \approx \frac{1}{2}$).

If $n^+ = n$, then all walkers are in a positive position and $\Delta s^+ = 0$. Over a long period of time, as the variance in walker position grows, through random chance at least one walker will eventually drift to a negative position, returning us to the case where $0 < n^+ < n$. If $n^+ = 0$, then all walkers are in a negative position, and $\Delta s^- > 0$. Over time, the walkers' average position grows more positive, until at least one becomes positive and we again re-enter the case $0 < n^+ < n$.

Altogether, we arrive at the following picture of the walker's dynamics. Walkers at a positive position drift with a negative rate down to zero, and walkers at a negative position drift with a positive rate up to zero. After sufficient time has elapsed, we would therefore expect the position of all walkers to be close to zero. However, for a walker s_u, positive updates increment the underlying coordinates ω_u^1 and ω_u^2 asymmetrically. Furthermore, a more positive coordinate receives a positive update with greater probability. Hence, we must have that $\omega_u^1 \gg \omega_u^2$ or $\omega_u^1 \ll \omega_u^2$. Because their sum must remain close to zero, it must be true that $\omega_u^1 \approx -\omega_u^2$. Thus, we have overall that $||\mathbf{v}_1|| = ||\mathbf{v}_2||$ and $\mathbf{v}^1 \cdot \mathbf{v}^2 / (\left|\left|\mathbf{v}^1\right|\right| \left|\left|\mathbf{v}^2\right|\right|) = -1$, confirming that a weight vector aligned with *different* examples adopts antiparallel components.

To validate our key assumption on μ, we simulate the Markov process 100 times with $L = 16$ and a batch size of 512. We empirically find that $\mu = 0.499 \pm 0.009$ (given by two standard deviations), matching closely our conjecture that $\mu = \frac{1}{2}$.

C.4.3 Magnitude of readouts The final piece to demonstrate in our study of the rich regime is that $|\overline{a^+}| < |\overline{a^-}|$, where $\overline{a^+}$ corresponds to the average across all positive readout weights and $\overline{a^-}$ corresponds to the average across all negative readout weights. Exactly characterizing these magnitudes is difficult, so we apply a heuristic argument based what we learned about the structure of parallel and antiparallel weights above and support it with numeric evidence.

Recall that the update rule for a readout weight a_i is given by

$$\Delta a = -\frac{c}{N}\sum_{j=1}^{N}\frac{\partial \mathcal{L}_j}{\partial f}\phi(\mathbf{w}\cdot\mathbf{x}_j),$$

where $c = \frac{\alpha}{\gamma\sqrt{d}}$ and α is the learning rate. Suppose $\mathbf{x}_j = (\mathbf{z}_u, \mathbf{z}_v)$. Then the update rule becomes

$$\Delta a = -\frac{c}{N}\sum_{u,v}\frac{\partial \mathcal{L}_{u,v}}{\partial f}\phi(\omega_u^1 + \omega_v^2).$$

If $\frac{\partial \mathcal{L}}{\partial f}$ is about the same in magnitude across all training examples, then our readout updates are proportional to

$$\Delta a \propto \sum_{u,v} S(u,v)\,\phi(\omega_u^1 + \omega_v^2).$$

where

$$S(u,v) = \begin{cases} 1 & u = v \\ -1 & u \neq v. \end{cases}$$

Let us first consider the case where $\mathbf{w}$ corresponds to a negative readout weight a^-. From above, we know that $\mathbf{w}$ is antiparallel. Hence, when encountering a same example, $\omega_u + \omega_u \approx 0$. If the magnitude of all coordinates are roughly equal, when encountering a different example, $\omega_u + \omega_v > 0$ about $1/4$ of the time. Comparing Eq (C.11) to Eq (C.16), we see that the expected drift for antiparallel weight vectors is roughly twice that of parallel weight vectors over long timescales. Altogether, since the number of *same* and *different* examples is balanced, we have overall that $|\Delta a^-| \propto 2\cdot\frac{1}{4} = \frac{1}{2}$ for $a^- < 0$.

Now consider the case where $\mathbf{w}$ corresponds to a positive readout weight a^+. From above, we know that $\mathbf{w}$ is parallel. For large batch sizes N, the expected drift of a walker at initialization is 0[7], so we expect $n^+ \approx n^-$. If the magnitude of all coordinates are roughly equal and L is large, when encountering a same example, $\omega_u + \omega_v > 0$ about $1/2$ of the time. When encountering a different example, $\omega_u + \omega_v > 0$ about $1/4$ of the time. Altogether, we would therefore expected $|\Delta a^+| \propto \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

From this rough estimate, we find that $\frac{|\Delta a^-|}{|\Delta a^+|} \approx 2$ for average negative and positive readouts. Since the rate of increase for negative readouts tends to be larger than that of positive readouts, we would expect the magnitude of negative readouts to be similarly larger. In fact, if readouts start with small initialization, we may conjecture that $\frac{|\overline{a^-}|}{|\overline{a^+}|} \approx 2$. In practice, this quantity turns out to be about 1.56 ± 0.09 (with 2 standard deviations), computed from 10 runs [8].

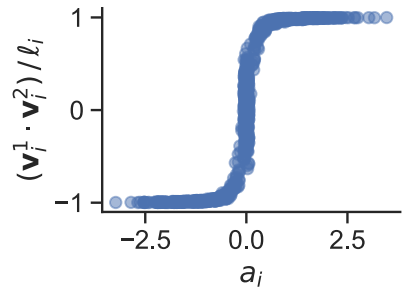

Figure C2: **Rich-regime weight structure when** $L = 2$. The model continues to develop parallel/antiparallel weights, though the magnitude of negative readouts is now about the same as the magnitude of positive readouts.

Note in the case that L is small (for instance, $L = 2$), our estimate $|\Delta a^+| \propto \frac{1}{4}$ breaks down since there may be only a single set of positive coordinates and no penalty is incurred on negative examples. In this case, we would have $|\Delta a^+| \propto \frac{1}{2}$, so $\frac{|\Delta a^-|}{|\Delta a^+|} = 1$. Indeed, this is exactly what we observe in the case where $L = 2$. Computing this quantity empirically yields 0.99 ± 0.07 (with 2 standard deviations), computed from 10 runs [9]. This outcome seems to be part of the reason why the model does not generalize well on the SD task with only 2 symbols, despite developing parallel/antiparallel weights (Figure C2). For $L \geq 3$, there seems to be sufficient pairs of positive coordinates in parallel weight vectors to restore the situation where $|\Delta a^-| > |\Delta a^+|$.

C.5 Test accuracy prediction

We apply our knowledge on the structure of $\mathbf{w}$ and a to estimate the test accuracy of the rich regime model. Our derivation is heuristic, but seeks to capture broad phenomena rather than achieve exact precision. We validate our predicted test accuracy in Figure C3, demonstrating excellent agreement.

Recall from Section 3.1 that a model achieving the handcrafted solution exhibits perfect classification of *same* examples. Any errors are therefore accumulated from misclassifying *different* examples. The crux of our estimate stems from approximating the classification accuracy of *different* examples.

Define I^+ to be the set of weight indices i such that $a_i > 0$, and define I^- to be the set of weight indices j where $a_j < 0$. Let $\mathbf{x}$ be a *different* example. Dropping constants that do not

[7] There is a 1/3 chance of observing a *same* example, which increments by 2. There is a 2/3 chance of observing a *different* example, which decrements by 1. Hence, the expected drift at initialization must be zero overall.

[8] The MLP has width $m = 1024$, and inputs have dimension $d = 512$. There are $L = 32$ training symbols

[9] As before, the MLP has width $m = 1024$, and inputs have dimension $d = 512$. There are $L = 2$ training symbols